SOC HOME LAB

SIEM-Based Attack Detection Using Splunk

SOC Portfolio Project · March 2026

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Certification	Certified SOC Analyst (CSA) – EC-Council
Project Type	SOC Home Lab – Portfolio Project
Target Role	SOC Analyst L1 / Cybersecurity Analyst
Tools Used	VMware · Kali Linux · Windows Server 2022 · Splunk
Date	March 2026
Status	Completed ✓

1. Project Overview

This project demonstrates the creation of a Security Operations Center (SOC) home lab used to simulate cyber attacks and detect malicious activity using a SIEM platform. The lab includes a Kali Linux attacker machine and a Windows Server 2022 target machine running inside VMware Workstation Pro 17. Splunk Enterprise was configured to collect and analyze Windows security logs in real time.

Key skills demonstrated in this project:

- Building and configuring a virtualized SOC lab environment
- Installing and configuring Splunk SIEM for Windows log collection
- Simulating real-world cyber attacks using Kali Linux tools
- Detecting and investigating security events using Splunk queries
- Mapping attack techniques to the MITRE ATT&CK; framework

2. Lab Architecture

The SOC home lab consists of two virtual machines connected through a NAT network inside VMware Workstation Pro 17.

Component	System	IP Address	Role
Attacker	Kali Linux 2025.2	192.168.52.x	Simulate cyber attacks
Target	Windows Server 2022	192.168.52.136	Generate security logs
SIEM	Splunk Enterprise 10.0.1	127.0.0.1:8000	Log monitoring & analysis
Virtualization	VMware Workstation Pro 17	Host Machine	Run virtual machines

Lab Flow

```
Kali Linux → Attack Simulation → Windows Server 2022 → Splunk SIEM → Log Investigation
```

3. Tools Used

Tool	Version	Purpose
VMware Workstation Pro	17	Virtual machine environment
Kali Linux	2025.2	Attacker machine – attack simulation
Windows Server 2022	Build 20348	Target machine – security log generation
Splunk Enterprise	10.0.1	SIEM – log collection and analysis
Nmap	7.95	Network scanning and port discovery
Hydra	9.5	Brute force login attack simulation

4. SIEM Setup – Splunk Configuration

Splunk Enterprise 10.0.1 was installed on the Windows Server 2022 machine and accessed via the web interface at **http://localhost:8000**. Windows local event logs were added as a data input to monitor the following sources:

- Security Logs – login events, authentication attempts, privilege use
- System Logs – operating system and service events
- Application Logs – application-level activity

Splunk dashboard after successful configuration:

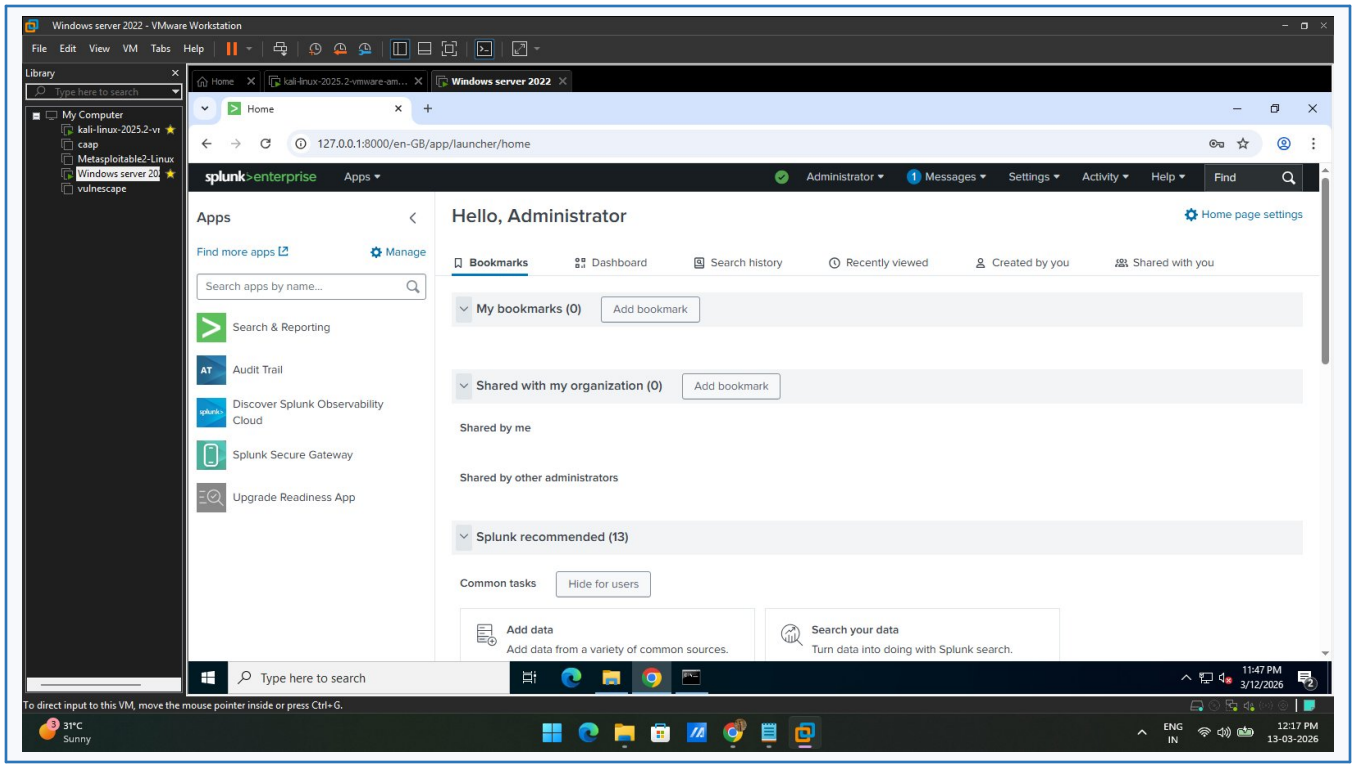

Figure 1: Splunk Enterprise dashboard running on Windows Server 2022 at http://localhost:8000

5. Network Connectivity Test

Before running attacks, network connectivity between the attacker machine (Kali Linux) and the target machine (Windows Server 2022) was verified. The Windows Firewall ICMP rule was enabled and a ping test confirmed successful communication.

```
ping 192.168.52.136
```

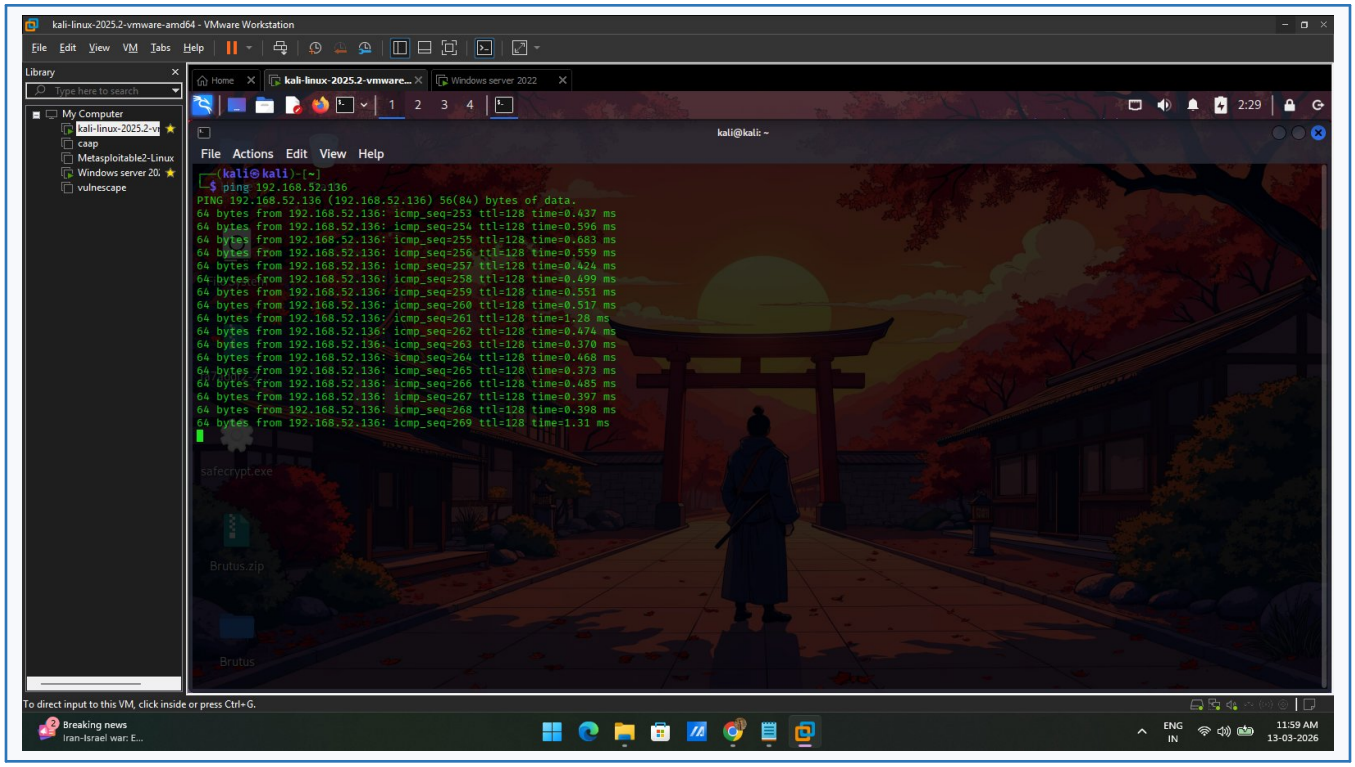

Figure 2: Successful ping replies from Windows Server 2022 (192.168.52.136) confirming network connectivity

6. Attack Simulation

6.1 Network Scanning — Nmap

A service version scan was performed from Kali Linux to identify open ports and running services on the Windows Server. This simulates the reconnaissance phase of a real-world attack.

MITRE ATT&CK | T1046 – Network Service Discovery

```
nmap -sV 192.168.52.136
```

Scan results:

- Host 192.168.52.136 – Status: UP (0.00041s latency)
- Port 5357/tcp – OPEN – Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
- Port 5985/tcp – OPEN – Windows Remote Management (WinRM)
- OS Detected: Windows | MAC: 00:0C:29:C4:07:62 (VMware)

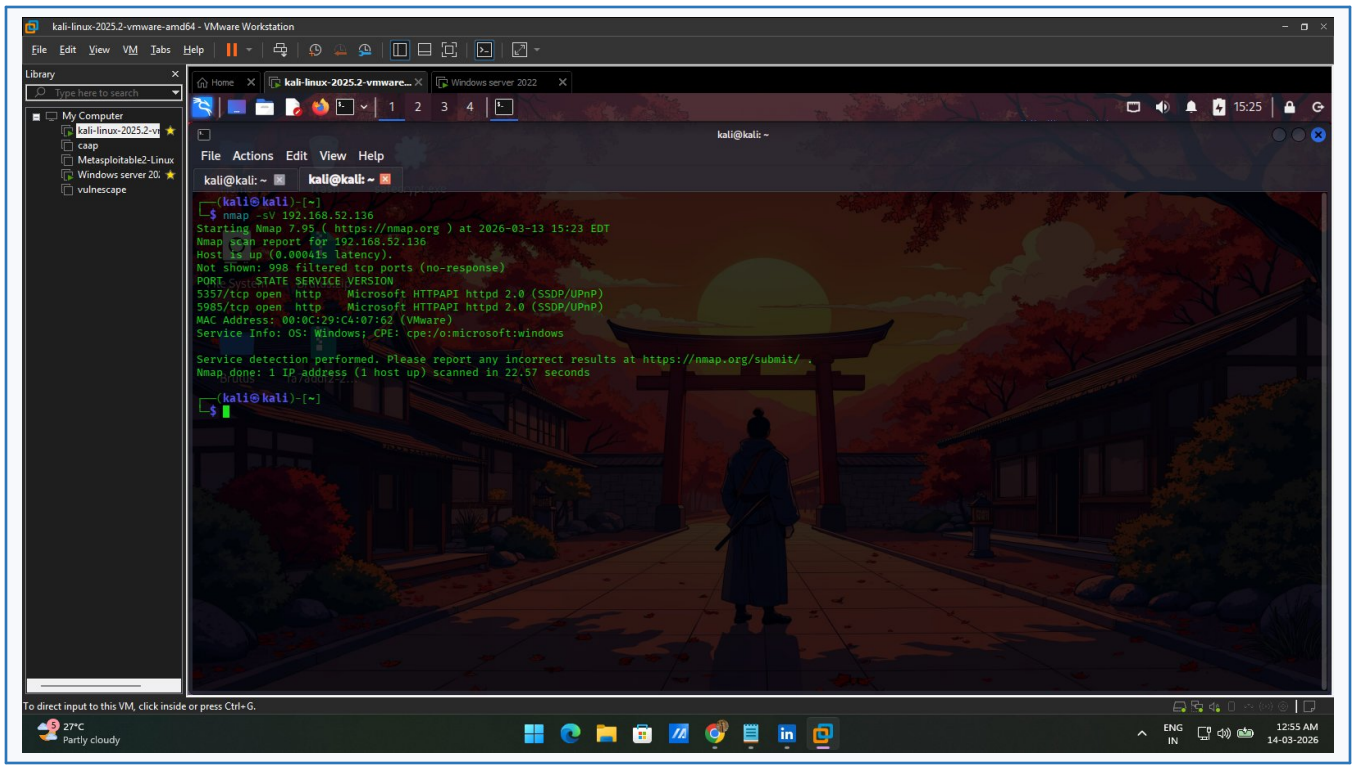

Figure 3: Nmap service scan from Kali Linux identifying open ports on Windows Server 2022

6.2 Brute Force Login Attack — Hydra

A brute force login attack was simulated using Hydra from the Kali Linux machine, targeting the SMB service (port 445) on the Windows Server. This generated multiple failed authentication events recorded in the Windows Security logs.

MITRE ATT&CK | T1110 – Brute Force / T1021.002 – SMB/Windows Admin Shares

```
hydra -l administrator -P /usr/share/wordlists/rockyou.txt 192.168.52.136 smb -V -f
```

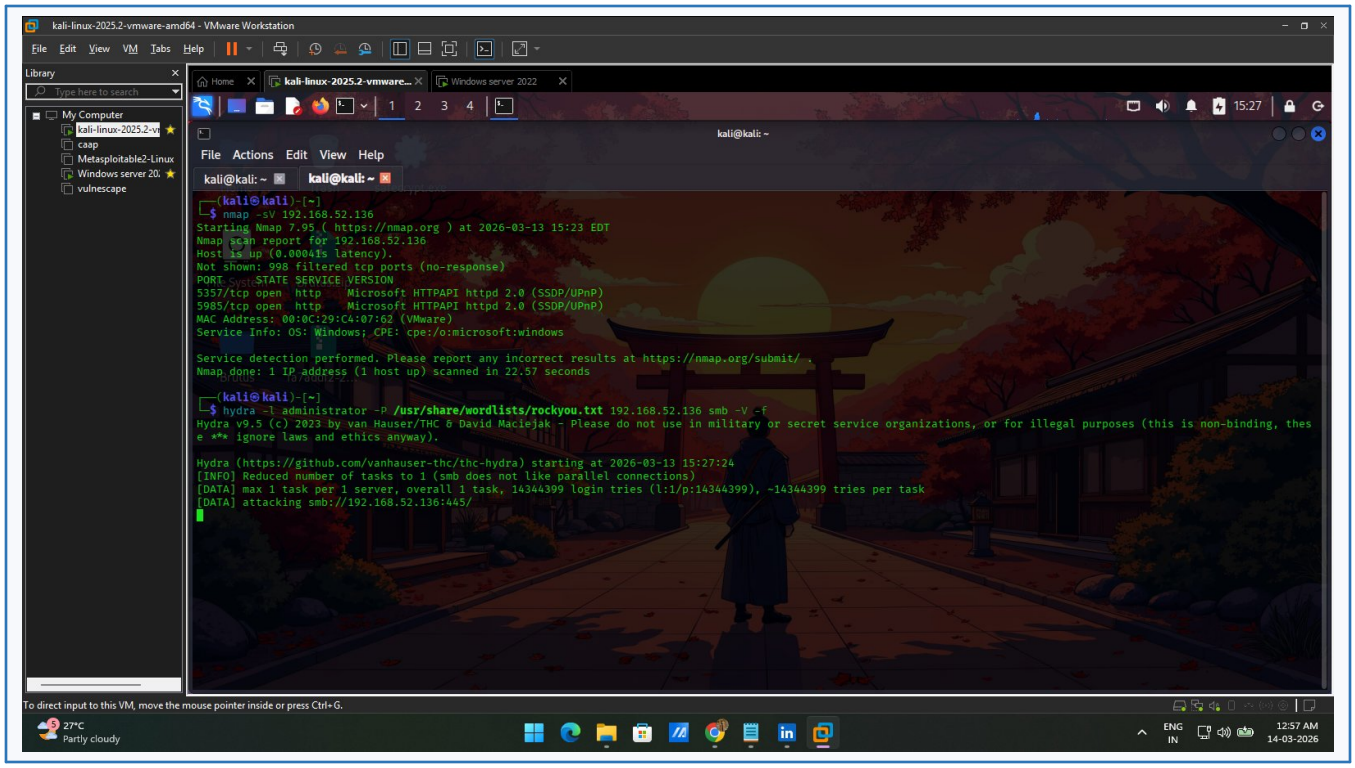

Figure 4: Hydra brute force attack running from Kali Linux targeting SMB port 445 on Windows Server 2022

7. Log Analysis & Investigation

7.1 Windows Event IDs Reference

Event ID	Event Name	Description
4624	Successful Logon	A user account successfully logged on to the system
4625	Failed Logon	Failed login attempt – key indicator of brute force attack
4688	Process Creation	A new process was created – monitors suspicious execution
4672	Special Privileges	Special privileges assigned – indicates privilege escalation

7.2 Splunk Detection – Failed Login Events

After running the brute force attack, Splunk was searched for **EventCode=4625 (Failed Logon)** events. The query returned **15 failed login events**, confirming the brute force attack was successfully detected and logged by the SIEM platform.

```
index=main EventCode=4625
```

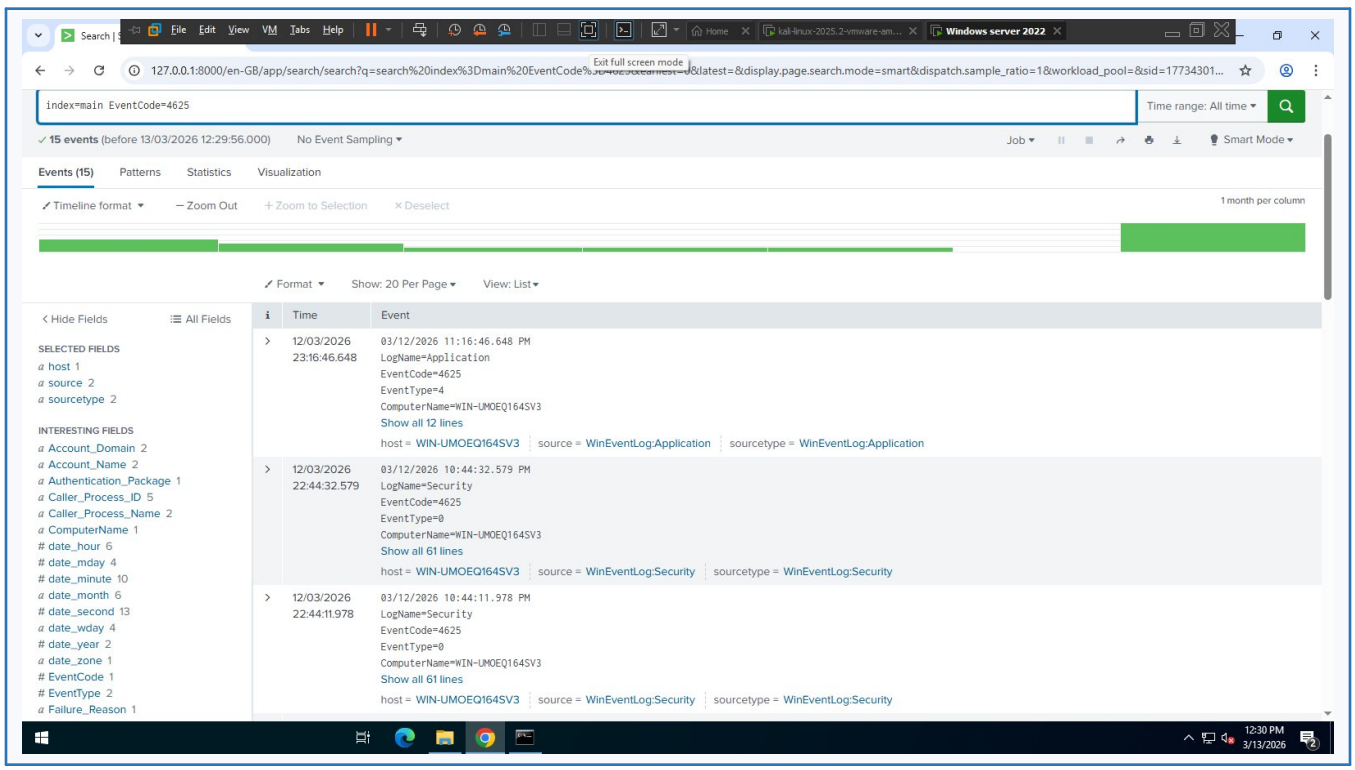

Figure 5: Splunk showing 15 EventCode=4625 (Failed Logon) events generated by the Hydra brute force attack

8. Findings

#	Finding	Evidence	MITRE ATT&CK;
1	Network Reconnaissance	Open ports 5357 & 5985 identified via Nmap scan	T1046
2	Brute Force Attack	15 EventCode=4625 failed login events in Splunk	T1110
3	SMB Service Targeted	Hydra attack against SMB authentication on port 445	T1021.002

9. Conclusion

This SOC home lab project successfully demonstrated how a SOC analyst monitors, detects, and investigates cyber attacks using a SIEM platform in a controlled lab environment.

Two real-world attack scenarios were simulated — network reconnaissance using Nmap and brute force login attempts using Hydra — and both were successfully detected and investigated using Splunk SIEM queries.

Skills demonstrated:

- Building and managing a virtualized SOC lab using VMware
- Configuring Splunk SIEM for real-time Windows log collection
- Simulating cyber attacks from Kali Linux (Nmap, Hydra)
- Detecting attacks using Splunk search queries and Windows Event IDs
- Mapping attack techniques to the MITRE ATT&CK; framework
- Documenting findings in a professional SOC investigation report

This project reflects the real workflow of a SOC analyst and serves as a portfolio project for entry-level cybersecurity roles.

10. Tools & References

- VMware Workstation Pro
- Kali Linux
- Windows Server 2022
- Splunk Enterprise
- Nmap
- Hydra
- MITRE ATT&CK; Framework